

Data Mining for Student Performance in Secondary Schools

CS 4412 - Data Mining | Final Report (M4)

Cesar Arevalo Colacho

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Abstract

This report presents a complete data mining analysis of 1,044 secondary school students drawn from two Portuguese schools. Using K-Means clustering as a baseline, supplemented by Principal Component Analysis, DBSCAN, Hierarchical Clustering, Decision Tree classification, and Local Outlier Factor anomaly detection, the analysis identifies three stable student performance profiles. Grade trajectory and failure history emerge as the dominant predictors of cluster membership, while parental education provides a secondary socioeconomic signal. All methods and findings are critically assessed for validity, limitations, and ethical implications.

1 Introduction

1.1 Dataset and Project Goals

This project applies data mining techniques to the UCI Student Performance Dataset (Cortez & Silva, 2008), which records academic and behavioral data from students enrolled in Mathematics and Portuguese Language courses at two secondary schools in Portugal. After merging the two course datasets and adding a course identifier attribute, the working dataset contains **1,044 students** and **34 attributes** spanning demographic background, family structure, study habits, and academic performance.

The central goal is to determine whether meaningful student groups can be discovered from this combination of attributes, and to identify which factors most strongly drive academic outcomes - moving beyond simple grade prediction toward a richer understanding of the patterns underlying student performance.

1.2 Discovery Questions

Three questions guide the entire analysis:

1. **Family and socioeconomic patterns:** Do parental education and family structure form recognizable patterns linked to academic consistency?

2. **Study habits and absences:** How do study time and absence counts interact to shape performance patterns?
3. **Natural student clusters:** What natural student groups emerge when combining demographic, family, and academic features?

2 Methods

2.1 Feature Selection

Eight variables were selected for clustering based on their correlation with the final grade (G3): **G1**, **G2**, **G3** (grade trajectory), **failures** (academic risk), **studytime** and **absences** (behavioral indicators), and **Medu** and **Fedu** (parental education). Nominal variables with arbitrary integer codes (school, job categories, reason, guardian, course) were excluded from distance-based algorithms to avoid introducing meaningless numeric distances between unordered categories. All eight selected variables were standardized to mean = 0 and SD = 1 using `scale()` before clustering.

2.2 Analytical Techniques

Five techniques were applied on top of the K-Means baseline, each serving a distinct analytical purpose:

Table 1: Analytical techniques applied and their primary roles.

Technique	Purpose	Key Parameter
K-Means Clustering	Primary clustering baseline	$k = 3$, $nstart = 25$
Principal Component Analysis	Redundancy reduction	3 components retained
DBSCAN	Density-based validation	$eps = 3.0$, $minPts = 5$
Hierarchical Clustering	Agglomerative validation	Ward's method, $k = 3$
Decision Tree	Cluster interpretability	$maxdepth = 4$, $cp = 0.01$
Local Outlier Factor (LOF)	Anomaly detection	$minPts = 10$, $threshold = 1.5$

The Elbow Method and Silhouette Analysis both independently converged on $k = 3$ as the optimal number of clusters. Silhouette scores were used throughout to compare method quality.

3 Results

3.1 Feature Relevance

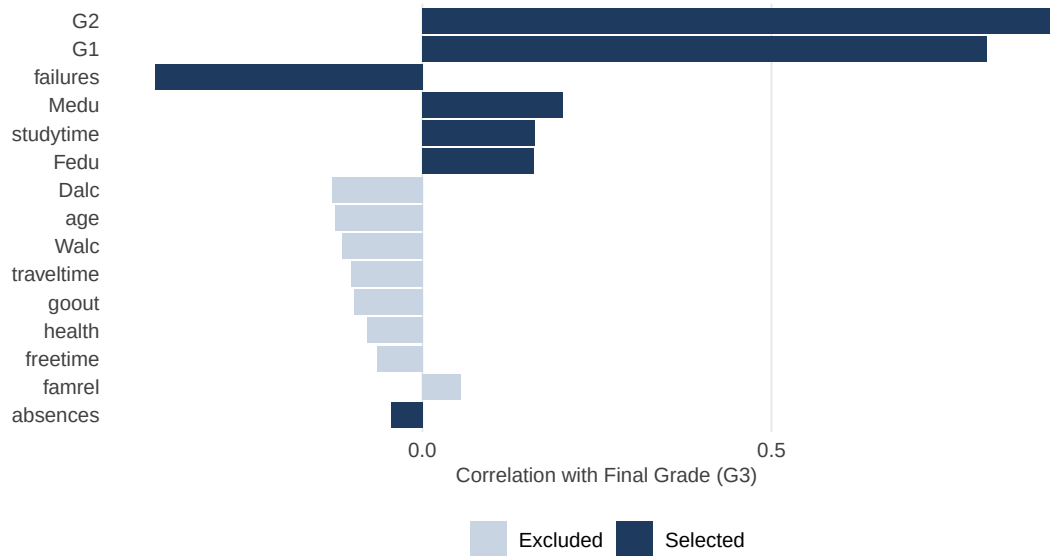


Figure 1: Correlation of each variable with final grade (G3). Blue bars identify the eight variables selected for clustering; gray bars were excluded due to near-zero correlation.

Figure 1 confirms that G1 and G2 (r approximately 0.81 and 0.91) are the strongest predictors of G3, followed by failures (r approximately -0.38) and the parental education variables Medu and Fedu (r approximately 0.15-0.20). Study time and absences contribute weakly but are retained to capture behavioral variation.

3.2 K-Means Cluster Profiles

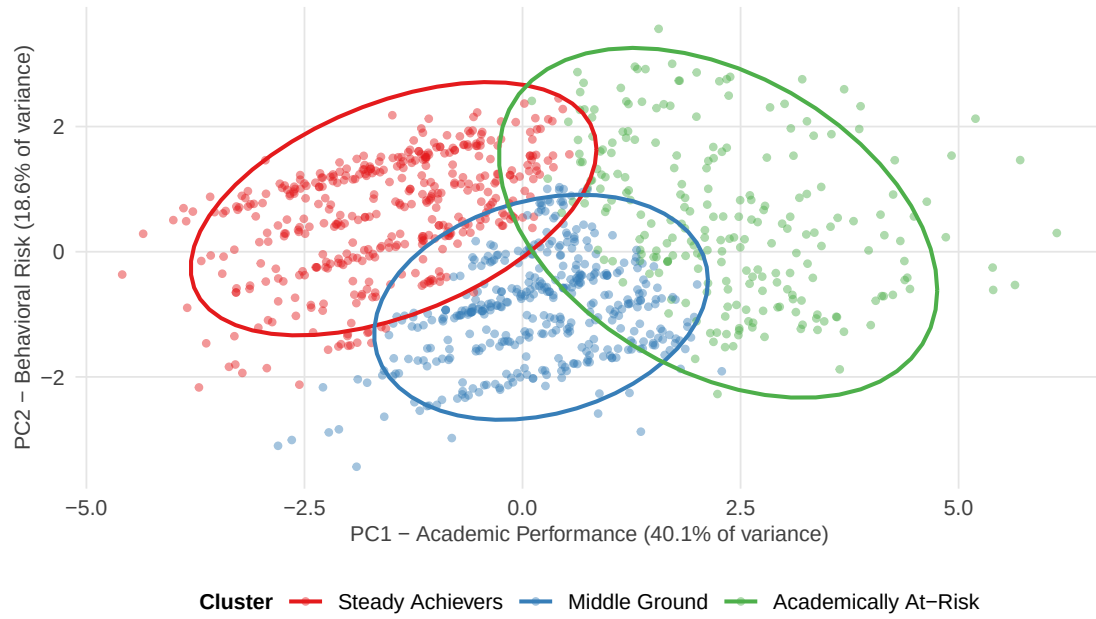


Figure 2: K-Means student clusters projected onto PCA space. PC1 captures overall academic performance (grade trajectory); PC2 captures behavioral risk and background factors. Ellipses show the 95% confidence region for each cluster.

K-Means identified three student profiles (average silhouette score = 0.1934). Table 2 summarizes the average feature values for each group.

Table 2: K-Means cluster profiles: average feature values per group.

Cluster	N	G1	G2	G3	Failures	StudyTime	Absences	Medu	Fedu
Steady Achievers	407	13.3	13.5	13.9	0.04	2.1	3.8	3.5	3.2
Middle Ground	401	11.2	11.3	11.7	0.07	2.0	3.7	1.8	1.7
Academically At-Risk	236	7.7	7.3	6.4	0.98	1.7	6.8	2.4	2.2

- **Cluster 1 - Steady Achievers:** Highest grades across all three periods, near-zero failure rate, and highest parental education levels. These students have both academic momentum and strong home support.
- **Cluster 2 - Middle Ground:** Average values on all features. The largest group, representing the typical student in the dataset.
- **Cluster 3 - Academically At-Risk:** Lowest grades and highest failure and absence rates. These students show the behavioral patterns most associated with academic struggle.

3.3 Method Comparison

All three clustering methods identified the same three-group structure. Table 3 compares their silhouette scores.

Table 3: Silhouette score comparison across clustering methods.

Method	Silhouette	Interpretation
K-Means (8 original features)	0.1934	Weak - gradual boundaries typical of education data
K-Means (3 PCA components)	0.2727	Weak-to-moderate - PCA reduces redundancy
Hierarchical Clustering (Ward, k = 3)	0.1410	Very weak - poor cluster cohesion

DBSCAN ($\text{eps} = 3.0$, $\text{minPts} = 5$) found a single large density region rather than three separate clusters. This is not a failure - it confirms that the student population forms a **continuous performance spectrum** rather than discrete density islands, which is consistent with the low silhouette scores across all methods. PCA improved the K-Means silhouette score from 0.1934 to 0.2727 by removing the grade redundancy (G1, G2, G3 correlate 0.81-0.91 with each other).

3.4 Decision Tree: Interpretable Rules

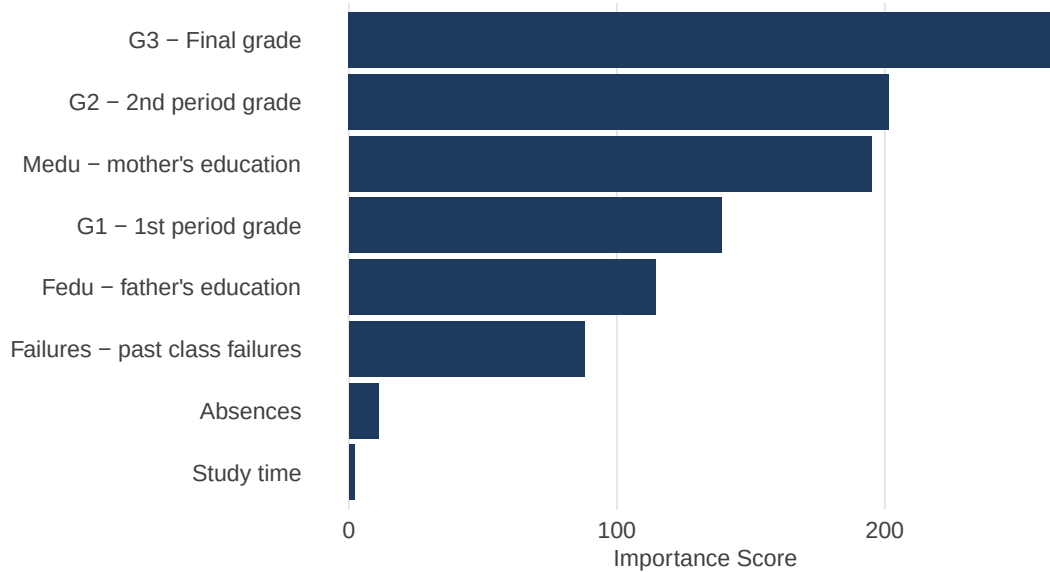


Figure 3: Feature importance scores from the Decision Tree trained to explain K-Means cluster labels. Higher scores indicate greater discriminative power between groups.

The decision tree achieved **88.7% accuracy** in assigning cluster labels using simple rules. The first split is on the second-period grade (G2), confirming that mid-year performance is the single most informative variable for separating all three groups. Failure history ranks second, followed by parental education. Study time and absences rank last, consistent with their low correlations with G3.

3.5 Anomaly Detection

LOF identified **16 anomalous students** (1.5% of the dataset). These students were distributed across all three clusters, not concentrated in one. Common anomaly profiles included students with very high absence counts but passing grades, and students with highly educated parents but a history of class failures - cases where the expected correlation between background and performance breaks down. These students represent situations where individualized attention is most warranted.

4 Interpretation

4.1 Discovery Question 1: Family and Socioeconomic Patterns

Family and socioeconomic factors form real but secondary patterns. Parental education (Medu, Fedu) correlates weakly with G3 (r approximately 0.15-0.20) and loads onto PC2 in the PCA, confirming a genuine but subordinate dimension. The decision tree uses parental education to differentiate moderate from high-performing students with otherwise similar grade trajectories. **Implication:** Schools cannot change a student's socioeconomic background, but identifying students with low parental education who are already showing grade decline offers an early actionable signal before the trajectory solidifies.

4.2 Discovery Question 2: Study Habits and Absences

Study time and absences both contribute to cluster membership, but they rank last among the eight clustering variables. LOF anomalies that included students with high absences and decent grades confirm that absence count alone is a poor early-warning signal. **Implication:** Monitoring absence trends in combination with grade trajectory and failure history provides a substantially more reliable risk indicator than tracking any single variable in isolation.

4.3 Discovery Question 3: Natural Student Clusters

Three natural student groups are consistently identified across K-Means, Hierarchical Clustering, and Decision Tree analysis. The cluster boundaries are **gradual**, not sharp - the student population forms a performance continuum rather than discrete categories. This is captured by the low silhouette scores and confirmed by DBSCAN's finding of a single dense region. **Implication:** The three profiles are most useful as a practical framework for designing tiered support strategies, not as fixed labels. Students near cluster boundaries, particularly those trending downward within the Middle Ground group, deserve close monitoring.

5 Critical Assessment

5.1 Validity

Silhouette scores of 0.14-0.27 fall in the “weak structure” range but are consistent with the nature of educational data, where performance is inherently continuous. The three-group finding is strengthened by its **consistency across methods** (K-Means, Hierarchical, and Decision Tree all agree), which reduces the probability that it is an artifact of any single algorithm’s assumptions. These clusters should be treated as descriptive approximations, not precise boundaries.

5.2 Limitations

Dataset scope: The data originates from two Portuguese secondary schools in 2008. The specific balance between parental education effects and grade trajectory may not transfer directly to other national systems, school types, or time periods. **Self-reported variables:** Study time, free time, and alcohol consumption are self-reported, introducing potential response bias that may suppress these variables’ true influence. **Snapshot limitation:** The dataset captures one academic year. A student’s trajectory changes over time; a single-year classification cannot substitute for longitudinal monitoring. **Feature selection bias:** Selecting variables by linear correlation with G3 may exclude non-linear or interaction effects.

5.3 Ethical Considerations

Labeling risk: Classifying students as “Academically At-Risk” using an unsupervised algorithm carries stigmatization risk. If shared with teachers, cluster labels could introduce confirmation bias, lowering expectations and creating self-fulfilling outcomes. **Equity:** Parental education is used as a clustering feature. This variable reflects socioeconomic inequality beyond any individual student’s control; using it to classify students risks encoding systemic disadvantage into institutional decisions. **Intended use:** These clusters are appropriate for system-level resource planning. They are **not** appropriate for individual-level academic decisions - placement, grading, or promotion - without direct human review and consideration of each student’s full context.

6 Conclusion

This analysis applied six data mining techniques to 1,044 secondary school students, consistently finding three performance profiles: Steady Achievers, Middle Ground, and Academically At-Risk. The primary finding is that **a student’s own academic history** - grade trajectory and failure count - is the dominant driver of cluster membership, while parental education contributes as a secondary socioeconomic signal. Study time and absences alone are insufficient early-warning indicators.

The multi-method approach strengthens confidence in the findings: K-Means and Hierarchical Clustering agree on three groups; PCA confirms that the core academic signal is robust after removing feature redundancy; DBSCAN’s finding of one large density region provides an important corrective - boundaries between groups are gradual, not sharp. The Decision Tree translates

this structure into transparent rules with 88.7% accuracy on cluster labels. LOF identifies 16 students whose profiles are genuinely unusual and warrant individual review.

Implications for practice: Schools seeking to use data-driven approaches should focus monitoring on grade trajectory declines and failure patterns, rather than absence counts in isolation. Cluster-based profiles are most valuable for designing differentiated support programs; they are least valuable - and carry the highest ethical risk - when applied as fixed individual labels. Any deployment should include transparency to students and families, human review of individual cases, and ongoing reassessment as student trajectories evolve.

Future work: Longitudinal data spanning multiple academic years would substantially improve the validity of these findings. Incorporating additional variables - teacher quality ratings, class size, school resources - would address the current dataset's most significant omissions. A cross-national replication study using the same attribute schema (facilitated by the survey template developed alongside this project) would test the generalizability of the three-cluster finding beyond the Portuguese secondary school context.

7 References

Cortez, P. and Silva, A. (2008). *Using Data Mining to Predict Secondary School Student Performance*. Proceedings of 5th Annual Future Business Technology Conference, Porto, Portugal, pp. 5-12.

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